

# Raspberry Pi 5 SSH Setup & Troubleshooting Guide (Windows 11 → Raspberry Pi)

This document describes how to connect from a Windows 11 PC to a Raspberry Pi 5 using SSH, along with troubleshooting steps and optional improvements.

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## 1. Network Setup Assumptions

- Raspberry Pi IP address: 192.168.4.70
  - Windows 11 PC IP address: 192.168.4.39
  - Both devices are on the same Wi-Fi network
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## 2. Step 1 — Verify Network Connectivity

On Windows (Command Prompt):

```
ping 192.168.4.70
```

Expected result:

- Replies received with low latency (e.g., 5–50 ms)
- 0% packet loss

If ping fails:

- Check Wi-Fi connection
  - Verify Raspberry Pi is powered on
  - Confirm IP address of the Pi
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## 3. Step 2 — Test SSH Port Access

On Windows PowerShell:

```
Test-NetConnection 192.168.4.70 -Port 22
```

Expected result:

```
TcpTestSucceeded : True
```

If False:

- SSH is not enabled or not running on the Pi

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## 4. Step 3 — Enable SSH on Raspberry Pi

On the Raspberry Pi (local terminal):

```
sudo systemctl status ssh
```

If not active, enable it:

```
sudo systemctl enable ssh  
sudo systemctl start ssh
```

Install SSH server if missing:

```
sudo apt update  
sudo apt install openssh-server
```

Verify listening port:

```
sudo ss -tlnp | grep :22
```

Expected:

```
LISTEN ... :22
```

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## 5. Step 4 — Connect via SSH from Windows

Basic connection:

```
ssh username@192.168.4.70
```

Example:

```
ssh wlabagnara@192.168.4.70
```

First-time connection prompt:

- Type `yes` to accept host key
- Enter Raspberry Pi password

Successful login looks like:

```
username@raspberrypi:~ $
```

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## 6. Common Issues & Fixes

### Connection timed out

- Wrong IP address
- Pi offline or not connected to Wi-Fi
- Firewall blocking connection

### Connection refused

- SSH service not running
- SSH not installed

### Permission denied

- Incorrect username or password
- Password authentication disabled

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## 7. Step 5 — Verify SSH Login Success

Once connected, you can run:

```
pwd  
ls -la  
hostname -I
```

Exit session:

```
exit
```

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## 8. Optional Improvement — SSH Shortcut (Windows)

Create file:

```
C:\Users\<your-user>\.ssh\config
```

Add:

```
Host rpi5
  HostName 192.168.4.70
  User wlabagnara
```

Now connect with:

```
ssh rpi5
```

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## 9. Optional Improvement — SSH Key Authentication

Generate key on Windows:

```
ssh-keygen
```

Copy key to Raspberry Pi:

```
type %env:USERPROFILE\.ssh\id_ed25519.pub | ssh wlabagnara@192.168.4.70 "mkdir -
p ~/.ssh && cat >> ~/.ssh/authorized_keys && chmod 700 ~/.ssh && chmod 600
~/.ssh/authorized_keys"
```

After this, login works without password:

```
ssh rpi5
```

## 10. Summary of Working Diagnosis Flow

1. `ping 192.168.4.70` → confirms network connectivity
  2. `Test-NetConnection -Port 22` → checks SSH port availability
  3. Enable/start SSH on Pi if needed
  4. `ssh username@192.168.4.70` → connect successfully
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